Predicting Human Reading Difficulty using Language Model Surprisal

Course: Computational Psycholinguistics **Team:** Mudit Gupta (2024201058)

Video — **PPT**

Understanding how humans process language in real time is central to computational psycholinguistics. The *surprisal* of a word—defined as the negative log probability of a word given its context—is a widely used predictor of processing difficulty in psycholinguistic research (Smith & Levy, 2013).
$S(w_i) = -\log P(w_i \mid \text{context})$

We evaluate whether surprisal computed from modern language models predicts human reading time and whether transformer-based models (GPT/BERT) outperform classical n-gram language models.

Research Questions & Hypotheses

- **RQ1:** Does language-model surprisal correlate with human reading time?
- **RQ2:** Do transformer LMs (GPT/BERT) predict reading difficulty better than n-gram LMs?
- **RQ3:** Do syntactic constructions (relative clauses, center-embedding, garden-path sentences) produce surprisal spikes and increased reading times?
- **Hypothesis:** Higher surprisal values will correspond to longer reading times. Transformer surprisal will explain more variance due to better modeling of long-range contextual dependencies.

Methodology

1. Surprisal Computation: Train baseline bigram and trigram language models using Kneser–Ney smoothing. Compute transformer surprisal: for autoregressive models use $-\log P(w_i \mid context)$; for masked models compute pseudo-log-likelihood and aggregate to word level.

2. Alignment and Preprocessing: Map model tokens to dataset word tokens by aggregating subword probabilities. Normalize punctuation and remove noisy or non-linguistic tokens.

3. Predictive Modeling: Fit linear and mixed-effects regression models (random intercepts/slopes for subject and item) as well as non-linear models such as Random Forest or XGBoost. Control variables include word length, frequency, part-of-speech, and sentence position.

4. Targeted Linguistic Analyses: Analyze surprisal spikes at syntactic disambiguation points (e.g., garden-path sentences). Such analyses have been widely used to study sentence-processing complexity in eye-tracking corpora (Demberg & Keller, 2008).

5. Limitations and Mitigation

- **Dataset heterogeneity:** Natural Stories uses self-paced reading while Dundee uses eye-tracking. We therefore analyze datasets separately and optionally normalize measures (log-RT and z-scores).
- **Tokenization mismatch:** Transformer models produce subword tokens. Word-level surprisal will be reconstructed by aggregating subword probabilities.
- **Participant variability:** Differences across readers can influence reading time. Mixed-effects models with subject random effects will control for this variability.
- **Model bias:** Language models trained on large corpora may encode distributional biases that do not perfectly reflect human cognition.

Datasets

Primary dataset: **Natural Stories Corpus** — a reading-time corpus designed to contain complex syntactic constructions (Futrell et al., 2018).

Replication dataset: **Dundee Corpus** — an eye-tracking corpus widely used to study predictability and sentence processing.

References

- Smith & Levy (2013) — Predictability effects on reading time.
- Demberg & Keller (2008) — Eye-tracking evidence for sentence processing complexity.